

Expense Tracker (Java CLI Application)

File-Based Expense Management in Java

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Project Overview

What This Project Is

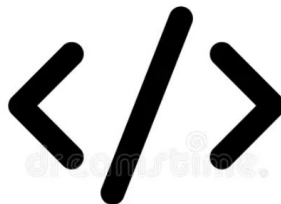
A simple, command-line Java application that allows users to record and analyze personal expenses.

Purpose

To demonstrate Java fundamentals, file handling, object-oriented design, and clean CLI interaction.

Key Requirements Met:

- Add and store expenses
- Calculate totals
- Summaries by category
- Trend analysis
- File-based persistence



Key Features

Key Features Included In This Project

- **Expense Management** — Add expenses, view all expenses
- **Analytics** — Total expense, category totals, highest/lowest category
- **Trends** — Date-sorted trend analysis
- **Storage** — Persistent file-based storage (data.txt)

```
==== Expense Tracker ====  
1. Add Expense  
2. View All Expenses  
3. Total Expense  
4. Expense by Category  
5. Expense Trend  
6. Highest & Lowest Category  
7. Exit  
□
```

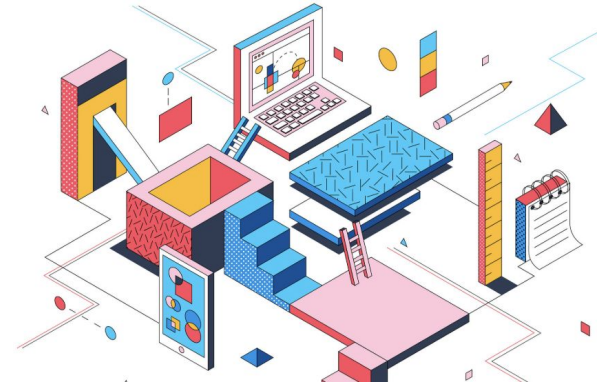
System Design

Core Components

- **Expense.java** – Represents a single expense
- **ExpenseManager.java** – Handles logic, calculations, and file I/O
- **Main.java** – CLI menu and user interaction
- **data.txt** – Stores all expenses in CSV format

Design Approach

- Object-Oriented Structure
- Separation of concerns
- Simple, readable, maintainable code



How The Application Works

A simple, file-based Java application that processes user input and persists data through a CLI workflow.

1. User selects an option from the CLI menu
2. Application reads/writes data to data.txt
3. ExpenseManager.java processes calculations
4. Results are displayed in the terminal
5. Data persists between runs

Technologies Used

- Java
- Local file storage
- Collections (List, Map)
- LocalDate API



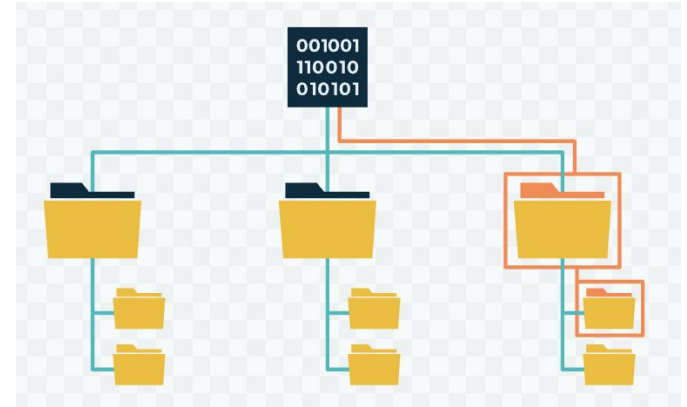
File Storage Format

Data is stored in a simple CSV format:

→ Category, amount, date

Example entries:

```
2026-05-01 | Food | $200.0
2026-05-01 | Food | $20.0
2026-05-01 | Travel | $50.0
2026-05-03 | Shopping | $100.0
2026-05-02 | Entertainment | $40.0
2022-12-09 | Food | $300.0
```



Demo Output

Sample CLI Output

→ Adding an expense

```
1
Category: Food
Amount: 120
Date (YYYY-MM-DD): 2026-05-03
Expense added!
```

→ Viewing all expense

```
2
2026-05-01 | Food | $200.0
2026-05-01 | Food | $20.0
2026-05-01 | Travel | $50.0
2026-05-03 | Shopping | $100.0
2026-05-02 | Entertainment | $40.0
2022-12-09 | Food | $300.0
2026-05-03 | Food | $120.0
```

→ Category totals

```
4
Travel : $50.0
Entertainment : $40.0
Shopping : $100.0
Food : $640.0
```

→ Highest and lowest categories

```
6
Highest: Food = 640.0
Lowest: Entertainment = 40.0
```

Conclusion

Summary

- Fully functional Java CLI application
- Clean object-oriented design
- Reliable file-based storage
- Meets all assignment requirements

Thank You!